



99-Million-Year-Old Fossil Flower

Scientists identify new fossil flower - *Valvuloculus pleristaminis*, once bloomed in the Cretaceous period. <http://digit.in/jan21-03>



Freshwater Skin Disease

A devastating skin disease afflicting dolphins worldwide has been linked to climate change. <http://digit.in/jan21-04>

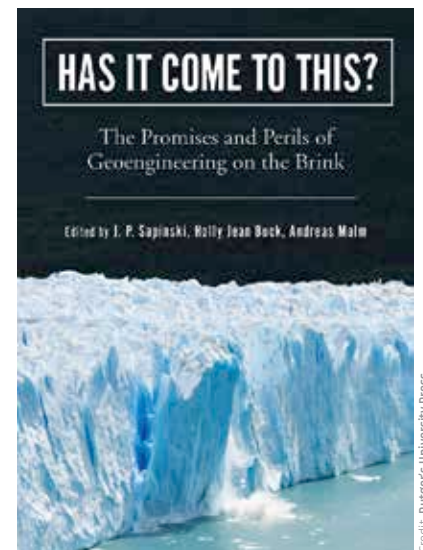
tion technologies. Across America and Europe, using electricity instead of gas for heating homes is further expected to reduce the reliance of civilization on fossil fuels. Finally, the increase in use of electricity for cooking across the world is expected to help. Even if all the countries come together and make the most drastic changes towards accelerated energy transition, it is still not going to be enough to meet climate change goals, specifically the Paris agreement. Even in the best case scenario, it is just too late to simply move to cleaner energy sources and call it a day. If the goals have to be met, then it is absolutely necessary to use at least some of the climate engineering technologies, at unprecedented scales. These might be outrageous, dangerous and untested ideas. Scientists have originally been thinking of them as a back up plan, an insurance policy in case all other approaches fail, but it now increasingly looks like we will have to be using some of these techniques because all other approaches have failed.

THE HUMAN IMPACT

When it comes to solar radiation management and carbon dioxide removal, the scientific community is divided on whether or not solar radiation management techniques should be con-

sidered and implemented at all. This is because artificially suppressing the temperature rise in the plant is at best a stop gap solution, and that there is really no alternative to removing the carbon dioxide in the atmosphere. The greenhouse gases in the atmosphere are still there, and as their chemistries continue to advance without changing course. For example, all the solar radiation management techniques in the world, implemented at large scales across the planet would still not be able to prevent the increasing acidification of the ocean caused by carbon dioxide in the atmosphere. Developing technologies to keep the temperature of the planet in check may be a viable short term alternative that becomes a trap that human civilization can fall into. In a scenario where there is widespread implementation of solar management, and at the same time industries take the introduction of these technologies as a license to pollute, the planet will be stuck in a situation where we are reigning in the temperature while continuing to pollute. This means that while some scientists support the idea of solar radiation management, other scientists are vehemently opposed to it. Perspectives from both sides of the issue are available in a new book that consists of a number of geoengineering related essays and interviews, called "Has It Come To This? The Promises and Perils of Geoengineering on the Brink". One of the contributors to the book, Holly Jean Buck explains exactly why solar management is controversial, "Solar geoengineering faces an enormous challenge relating to how it might be governed, since it's a planetary-scale intervention. The politics are very complicated. Carbon removal is something that can be implemented all around the world in different ways, but solar geoengineering is something that you'd want a global consensus on. I could very easily see a situation where a couple of countries agree to do it, and nobody disagrees with them, which is different from a global consensus"

Then there are the carbon dioxide removal techniques, that are not without their own issues. The technologies closest to implementation is called direct air capture and is based on the technology developed by the European Space Agency in the 1990s, to remove carbon dioxide from space stations and space ships. These technologies involve using fans to blow air through filters that remove the



Credit: Rutgers University Press

carbon dioxide. There are also artificial trees being developed for sequestering carbon, with proposals to line these trees along highways where the emissions are the most. The direct on ground impact of such technologies is expected to be on the lines of massive infrastructure projects where communities are displaced, such as dams. Also there is the pressing question of who will bear the cost of these projects. The outlook is looking dire indeed because we cannot use clean fuels quickly enough, managing planetary heat could lead us to a trap, removing carbon dioxide uses expensive and untested technologies. All of these together are not enough to contain the changes that industrial activity has introduced to the planet. More detailed information about specific carbon dioxide removal and solar radiation management techniques can be found in the September 2018 issue of Digit. digit.in



The aim of ocean fertilization techniques is to use iron pellets to simulate phytoplankton blooms, like this one off the coast of Argentina.